

Exploring the Relationship Between C60 Deposition Cycles and UV EQE Degradation

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How does the number of C60 deposition cycles used when manufacturing a cell relate to its EQE degradation over the UV band?

Methodology:

- Calculate mean EQE across the 350-400nm wavelength band for every week of data for each pixel.
- Create max-normalized EQE time series for each pixel with at least 5 weeks of measurements.
- Separate series by C60 condition and compute weekly mean EQEs and SEMs for each distribution.
- Uniformly resample weekly mean EQE within each week's SEM to get 10,000 bootstrapped time series for each distribution.
- Perform a quadratic fit on each bootstrapped time series and compare quadratic and linear coefficient distributions across C60 condition using ANOVA.

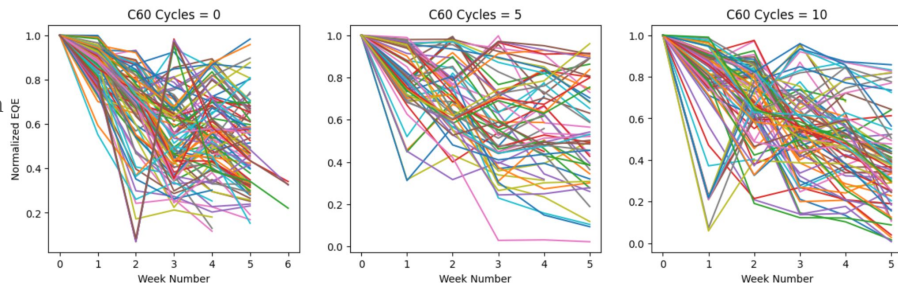
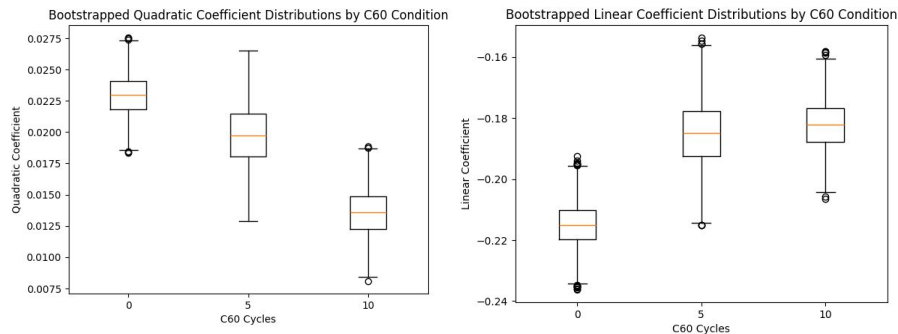


Figure 1: Normalized EQE time series by C60 condition, excluding series whose baseline measurements were not their maxes.

Figure 2: Bootstrapped quadratic (left) and linear (right) coefficient distributions for quadratic fit of bootstrapped EQE time series, across C60 condition.



Conclusion: Both ANOVAs yield p values below a 0.01 significance level, implying that the number of C60 deposition cycles has a highly statistically significant effect on the average quadratic and linear coefficients describing EQE degradation over time. The bootstrapped quadratic coefficient distributions (Figure 2) shift downward with increasing C60 cycle count, indicating that higher numbers of cycles are correlated with reduced curvature or acceleration of EQE degradation. Similarly though less strongly, we observe linear coefficients becoming less negative with more cycles, indicating weaker linear decay.